EBU5403 Alternative Assessment (Odd Semester - Paper A)

Joint Programme Assessments 2022/23

EBU5403 Internet Protocols

Answering this paper requires **2 hours**; Answers to be submitted within the allocated **3 hours window**.

Answer ALL questions

INSTRUCTIONS

1. **You must NOT share any content from this document during the assessment period.**
2. Your answers must be typed, and diagrams or equations must be written clearly and legibly with black or blue colour **and in English**.
3. You need to submit your answers BEFORE the allocated deadline.
4. **Read the instructions on the inside cover of the questions sheet.**

Examiners

Dr Richard Clegg, Dr Xianhui Che

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Filename: EBU5403_ALT-2022-23_S1A

Instructions

This is an open-book assessment, which should be completed **within 2 hours**. You MUST submit your answers within 3 hours from the assessment being released.

You MUST complete the assessment on your own, without consulting any other person. You MAY NOT check your answers with any other person.

You can refer to textbooks, notes and online materials to facilitate your working, if you provide a direct quote, or copy a diagram or chart, you must cite the source.

Before you start the assessment

1) Read the questions thoroughly and understand them.

2) Ensure you have all the resources you require to complete and upload the final assessment.

3) If you require any assistance, **raise the issue via the messaging section of this assessment on QMPlus**, immediately.

During the assessment session

1) Use the supplied answer sheet document to enter your answers. Start on a new page for each question. Make sure it is clear which question number you are answering.

2) **Type your answers** in the supplied answer sheet; hand-written equations or sketches can be incorporated into the answer sheet. Please save your work at least every 15 minutes so that you do not risk losing it.

3) When completed answering all questions, perform a word count and list the number of words on the answer sheet, then save the file as pdf before uploading, **only pdf will be accepted**, any other file format will not be accepted.

4) Your submission must be your own work, and you must ensure that you do not break any of the rules in the Academic Misconduct Policy.

Submitting the Assessment

1) You will have 3 hours from the start of the scheduled assessment time – do not leave submissions too close to the deadline. NO late submission will be accepted, no exceptions.

2) Make sure you upload and submit the final version before the deadline.

3) Please be aware that submissions will be subject to review, including but not limited to plagiarism detection software.

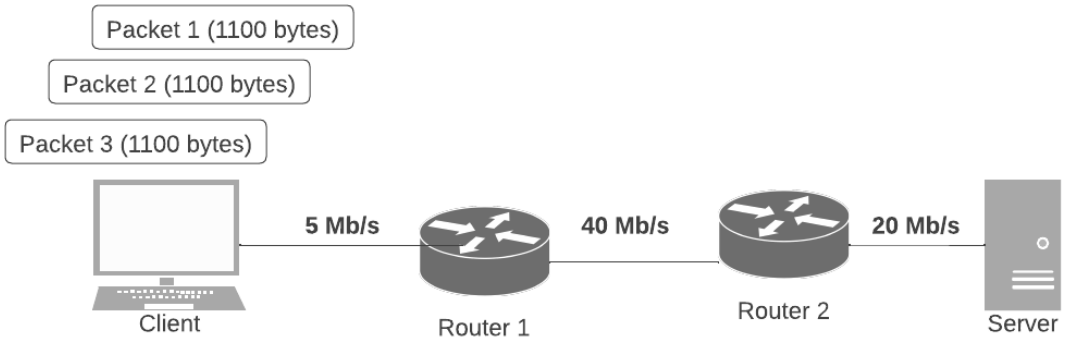
If you have any problems relating to access or submitting during the assessment period, please contact the email (it-issues@qmbupt.org), state the module code in the subject, and clearly state your name and student ID and any issues you are experiencing. You must use either @qmul.ac.uk or @bupt.edu.cn email address. Requests from external email addresses will not be processed.

Question 1

a) This question tests your understanding of Internet performance.

Consider a network shown in the figure and answer the questions below:

[8 marks]



i) Assuming all the three packets are transmitted in order immediately one after another, how long will it take for the third packet to arrive at the server? You can ignore the nodal processing delay and propagation delay. (Your answer must include the calculation process.)

(4 marks)

ii) Calculate the round trip time (RTT) between the client and the server. Give an example of the network command displays RTT information.

(2 marks)

iii) Identify the bottleneck link and comment on how it may impact the network.

(2 marks)

b) This question tests your understanding about network structure.

Consider a story described below and answer the questions below:

John lives in Shahe district in Beijing. One day he came home and found that he had no WiFi signal. He checked his **WiFi router** and called China Telecom where he got his broadband from. China Telecom informed him that one of their **access routers** in Shahe district was down, and assured him all their major **city routers** connecting Beijing and other provinces were still in good working order. John had no choice but to use his **smartphone** as a **hot spot** to connect to the Internet, but it cost too much, so instead he went to a computer lab in BUPT campus with 30 computers connected by a **switch**. He downloaded some files from a **BUPT server**, and then accessed the **CERNET** (Chinese Education Internet) via the **BUPT gateway**. He tried to use Google but failed because of national policies enabled on **backbone routers**.

[11 marks]

i) Categorise the highlighted words into two groups: edge network and core network.

(5 marks)

ii) Using OSI model as a reference model, name all the layers that exist in the following devices:

- John's smartphone
- Backbone router
- The switch in the computer lab

(6 marks)

c) This question tests your understanding about the Transport layer.

[6 marks]

i) Explain the purpose of port numbers in TCP or UDP.

(2 marks)

ii) Recommend a respective transport protocol to support the following services and **briefly** give your reasons:

- Sending an email
- Voice over IP (real time telephone calls)

(4 marks)

Question 2

a) This question is about setting the TCP timeout.

[9 marks]

i) What problem can be caused by setting the TCP timeout too short?

(2 marks)

ii) What problem can be caused by setting the TCP timeout too long?

(2 marks)

iii) If you know that your network has quite stable RTT but suffers a lot of packet loss, would you prefer a short TCP timeout or a long one? **Briefly** say why this is.

(3 marks)

iv) In estimating RTT we use:
EstimatedRTT = (1- α)*EstimatedRTT + α*SampleRTT

with a typical value for α of 0.125. Give **one** advantage and **one** disadvantage of setting a higher value.

(2 marks)

b) This question is about subnets. Your company has a class B subnet 132.0.0.0/16. You want to divide it as efficiently as possible assigning no more space than you currently need and assigning the largest networks first. You are breaking it into the following networks using variable length subnet masks.
A) 10000 hosts

B) 7000 hosts

C) 253 hosts

[10 marks]

i) Give the address range for each subnet A, B, C.

(6 marks)

ii) Give the broadcast address for each subnet.

(1 mark)

iii) If you need to assign a new subnet D without moving A,B,C what range will you use if it is a /24 network?

(1 mark)

iv) If you need to assign a new subnet D without moving A,B,C what range will you use if it is a /17 network? What problem might this cause in the future?

(2 marks)

c) Pipelined protocols: Consider Go-back-N and Selective Repeat. You are designing a new network that must use one of these two protocols.

[6 marks]

i) A network has high packet loss. You want to choose a protocol which is efficient at increasing throughput. Which of the two protocols do you pick? **Briefly** describe why.

(3 marks)

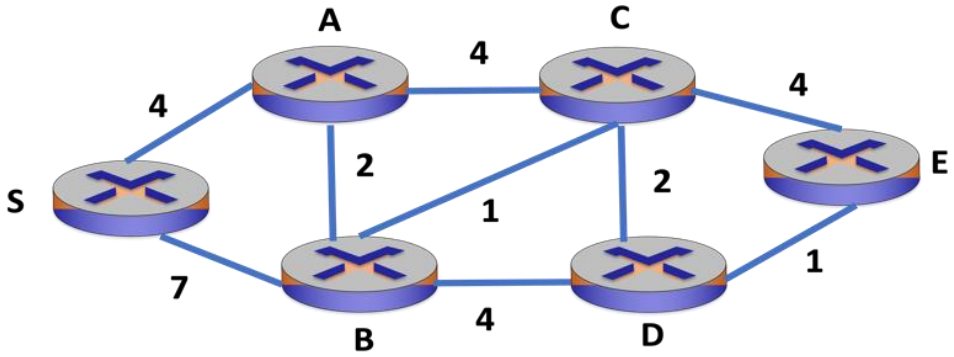
ii) The computer sending to the network has very little processing power and hence must maintain little state for each network connection. Which of the two protocols do you pick? **Briefly** describe why.

(3 marks)

Question 3

a) Consider the network below. Routes are to be found using Dijkstra's algorithm beginning at S. Suppose, N' is the set of nodes where the distance is known, D(x) is the currently estimated distance to node x and p(x) is the node prior to x on the current iteration.

[11 marks]



i) Draw a table showing N', D(x) and p(x) at each step of the algorithm.

(9 marks)

ii) Draw the resulting shortest path from S to E.

(2 marks)

b) This question is about Internet Protocol version 6.

[5 marks]

i) Give **three** main motivations behind IPv6?

(3 marks)

ii) **Briefly** explain tunnelling within the context of IPv4 to IPv6 transition.

(2 marks)

c) This question is about SDN**.**

[9 marks]

i) The SDN controller is in the control plane. **Briefly** explain what it does.

(4 marks)

ii) Name the two protocols in the SDN controller. What are their functions?

(3 marks)

iii) Where is the equivalent of the forwarding table generated in SDN compared with the traditional model?

(2 marks)

Question 4

a) This question is about multiple access technologies.

[10 marks]

i) **Briefly** describe in ALOHA what is meant by the terms **slotted** and **unslotted**?

(4 marks)

ii) Give **one** advantage and **one** disadvantage of slotted ALOHA.

(2 marks)

iii) Give **three** advantages and **one** disadvantage of simple time division multiple access when you have a fixed number of stations?

(4 marks)

b) Consider three LANs interconnected by two routers as shown in Figure Q4.1.

[10 marks]

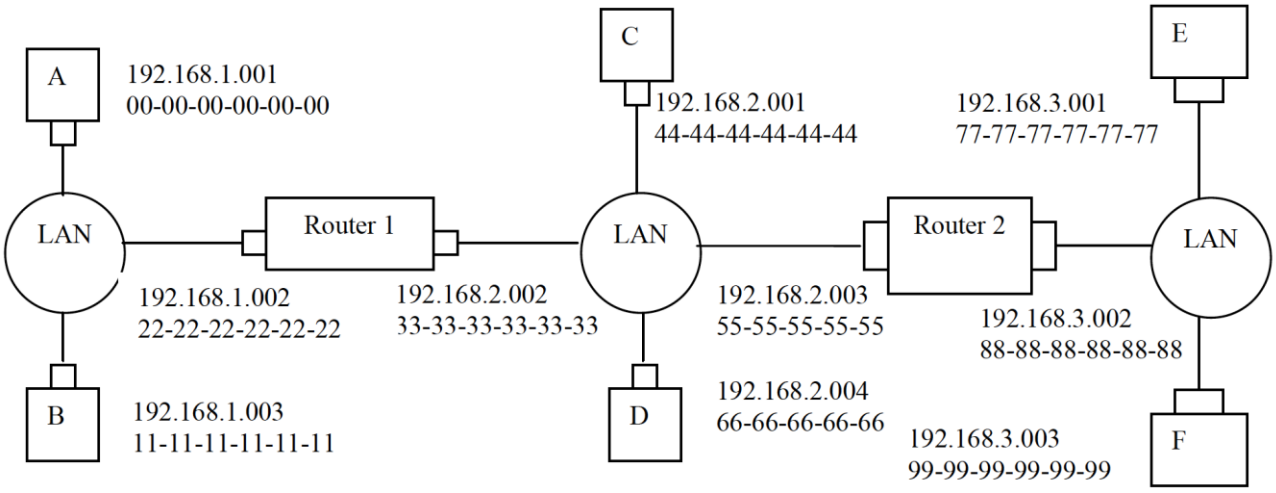


Figure Q4.1

i) Assuming all the ARP tables are up to date, provide all the steps required to send an IP datagram from Host E to Host B.

(6 marks)

ii) Repeat question b(i), now assuming that the ARP table in the sending host is empty and all other tables are up to date.

(2 marks)

iii) An ARP query is sent within a broadcast frame, and the ARP response is sent within a frame with a specific destination MAC address. Is this statement True or False. Indicate your answer and provide your explanation.

(2 marks)

c) An IEEE 802.3 CSMA/CD network is designed with a maximum one-way propagation delay of 1ms and a bit-rate of 100Mbps. Explain the need for a minimum frame size and calculate what it is in this case.

[5 marks]

END OF PAPER

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